

The Effects of Public Access Defibrillation on Survival After Out-of-Hospital Cardiac Arrest

A Systematic Review of Observational Studies

BACKGROUND: Despite recent advances, the average survival after out-of-hospital cardiac arrest (OHCA) remains <10%. Early defibrillation by an automated external defibrillator is the most important intervention for patients with OHCA, showing survival proportions >50%. Accordingly, placement of automated external defibrillators in the community as part of a public access defibrillation program (PAD) is recommended by international guidelines. However, different strategies have been proposed on how exactly to increase and make use of publicly available automated external defibrillators. This systematic review aimed to evaluate the effect of PAD and the different PAD strategies on survival after OHCA.

METHODS: PubMed, Embase, and the Cochrane Library were systematically searched on August 31, 2015 for observational studies reporting survival to hospital discharge in OHCA patients where an automated external defibrillator had been used by nonemergency medical services. PAD was divided into 3 groups according to who applied the defibrillator: nondispatched lay first responders, professional first responders (firefighters/police) dispatched by the Emergency Medical Dispatch Center (EMDC), or lay first responders dispatched by the EMDC.

RESULTS: A total of 41 studies were included; 18 reported PAD by nondispatched lay first responders, 20 reported PAD by EMDC-dispatched professional first responders (firefighters/police), and 3 reported both. We identified no qualified studies reporting survival after PAD by EMDC-dispatched lay first responders. The overall survival to hospital discharge after OHCA treated with PAD showed a median survival of 40.0% (range, 9.1–83.3). Defibrillation by nondispatched lay first responders was associated with the highest survival with a median survival of 53.0% (range, 26.0–72.0), whereas defibrillation by EMDC-dispatched professional first responders (firefighters/police) was associated with a median survival of 28.6% (range, 9.0–76.0). A meta-analysis of the different survival outcomes could not be performed because of the large heterogeneity of the included studies.

CONCLUSIONS: This systematic review showed a median overall survival of 40% for patients with OHCA treated by PAD. Defibrillation by nondispatched lay first responders was found to correlate with the highest impact on survival in comparison with EMDC-dispatched professional first responders. PAD by EMDC-dispatched lay first responders could be a promising strategy, but evidence is lacking.

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Sources of Funding, see page 963

Key Words: defibrillators
■ emergencies ■ heart arrest
■ out-of-hospital cardiac arrest

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Clinical Perspective

What Is New?

- Current guidelines on resuscitation recommend public access defibrillation (PAD) programs to improve the survival from out-of-hospital cardiac arrest (OHCA), but do not distinguish between different types of PAD strategies.
- This systematic review examined the effect on OHCA survival according to type of PAD strategy.
- Our analysis showed a median overall survival to hospital discharge after OHCA treated with PAD of 40.0% (range, 9.1–83.3).
- Defibrillation by nondispatched lay first responders was associated with the highest survival with a median of 53.0% (range, 26.0–72.0), whereas defibrillation by Emergency Medical Dispatch Center–dispatched professional first responders (firefighters/police) was associated with a median survival of 28.6% (range, 9.0–76.0).

What Are the Clinical Implications?

- This systematic review shows that early defibrillation by nondispatched lay first responders or Emergency Medical Dispatch Center–dispatched professional first responders (firefighters/police) correlates with increased survival after OHCA.
- Nondispatched lay first responders seem to have the potential to save the highest proportion of patients with OHCA.
- Focus on engaging laypersons to either use an on-site automated external defibrillator or dispatch them via the dispatch center should be prioritized, because not only can they reach the patient with OHCA quicker, but they can also reach patients with OHCA in residential areas.
- New PAD strategies with Emergency Medical Dispatch Center–dispatched lay first responders could be a promising strategy, but evidence is lacking.

Each year, >350 000 individuals in North America and 300 000 individuals in Europe have an out-of-hospital cardiac arrest (OHCA).^{1,2} Despite efforts to improve the outcome, the average survival after adult OHCA remains <10%.^{3–5}

As time passes, the number of patients with a shockable rhythm decreases,^{3,6,7} and as a result, the chance of survival diminishes with 7% to 10% for every minute that passes without intervention.⁶ Early defibrillation, preferably before emergency medical services (EMS) arrival, is therefore a crucial predictor for OHCA survival.^{3,7,8} Placing automated external defibrillators (AEDs) in the community as part of a public access defibrillation program (PAD) has shown survivals between 50% and 70%.^{9–14} As a result, PAD programs are recommended by the 2015 guidelines from the Ameri-

can Heart Association and the European Resuscitation Council.^{8,15} In agreement with these recommendations, national PAD programs have been implemented in several countries,^{16–20} and numerous states in the United States have made AED availability mandatory in certain public places.²¹

Despite the increasing availability and recommendations of PAD, <5% of all patients with OHCA are defibrillated before the arrival of EMS.^{7,22} Because time to defibrillation is crucial for improved survival, early defibrillation is the primary concern, whereas who brings the AED is of lesser importance. Several strategies have been proposed on how to increase and make the most of PAD. At present, 3 strategies exist: nondispatched lay first responders (on-site AED), professional first responders such as firefighter or police (mobile-AED) dispatched by the emergency medical dispatch center (EMDC), or EMDC-dispatched lay first responders (on-site or mobile AED).

Whereas current guidelines on resuscitation recommend PAD programs,^{8,15} the effect on OHCA survival according to type of PAD strategy has not previously been examined. We therefore performed a systematic review with the aim to evaluate the effect of 3 different PAD strategies on OHCA survival to hospital discharge.

METHODS

The systematic review is reported in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines,²³ which is endorsed by numerous journals.²⁴ The protocol for this study is registered in the PROSPERO database of systematic reviews (registration number: CRD42014013468).²⁵

Systematic reviews do not require approval by institutional review committees, nor do study subjects need to consent and this was therefore not sought.

Eligibility Criteria

We developed the research question following the PICOS (population, intervention, comparison, outcome, and study design) format, as recommended by the Cochrane Collaboration.²⁶ We identified studies reporting results from patients with OHCA (=population) with AED attachment or defibrillation by nonmedical professionals before the arrival of the EMS (=intervention). We did not mandate that studies include a control group. We included studies reporting survival to hospital discharge or 30-day survival reported in percentages (=outcome). To ensure comparability between studies, only observational studies, written in English, reporting OHCA of >10 patients were accepted (=study design). We excluded letters, case reports, abstracts, conference papers, and reviews. Studies including only patients <18 years of age were also excluded. PAD was divided into 3 subgroups according to who had deployed the defibrillator: nondispatched lay first responders (laypersons initiating cardiopulmonary resuscitation [CPR] without direction of the EMDC and deploying an on-site AED), professional first responders dispatched by the

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((((((((heart arrest[MeSH Terms]) OR "cardiac arrest") OR "cardiac arrests") OR "heart arrest") OR
"OHCA") OR "OOHCA") OR "OH-CA") OR "OOH-CA") OR "Sudden cardiac death")) AND
((((((((defibrillators[MeSH Terms]) OR "Automated external defibrillator") OR "AED") OR "Automatic
external defibrillators") OR "Public access defibrillation") OR "PAD") OR "Defibrillation") OR "first-
responder") OR "Automatic external defibrillator")) AND (((("Survival") OR "ROSC") OR "Return of
spontaneous circulation") OR "Cerebral performance category") OR "overall performance categories")

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Figure 1. Full electronic search strategy from PubMed.

AED indicates automated external defibrillator; OHCA, out-of-hospital cardiac arrest; PAD, public access defibrillation; and ROSC, return of spontaneous circulation.

EMDC (firefighters or police equipped with an AED), or lay first responders dispatched by the EMDC (laypersons directed to an OHCA with the help of the EMDC, eg, via a text-message alert system or smartphone-based system, and using either an on-site or mobile AED). Whenever it was not transparent whether the AED had only been attached or had been attached and defibrillated, we categorized it as only attached.

Information Sources and Search Methods

We conducted a literature search in PubMed, Embase, and the Cochrane Library on August 31, 2015. The full electronic search strategy can be seen in Figure 1.

The search strategy was further modified to fit Embase and the Cochrane Library.

Study Selection

Two reviewers (S.V. and J.S.B.) independently screened titles and abstracts. Interrater reliability was calculated using

Cohen's κ statistics. The same reviewers independently screened full texts of the selected studies. Here, disagreement was resolved by discussions until consensus. Data extraction and assessment of risk of bias were performed by the first author (J.S.B.) and confirmed by a second author (S.V.). If patient data overlapped between publications, they were considered as duplicate publications. The publication covering the longest time period was selected and the other(s) excluded.

To ensure capture of all important studies, all reference lists of the screened full-text studies were evaluated by the first author and judged whether to pursue further.

Data Collection Process and Data Items

Data were extracted into predetermined tables by the first author (J.S.B.). Data on study characteristics included study period, study duration, geographical location, study design, name of registry, and the Newcastle Ottawa Scale score. Data on AED circumstances and outcomes included proportion of CPR, proportion of witnessed OHCA, AED provider (nondispatched lay first responders, EMDC-dispatched professional first responders [firefighters/police], or EMDC-dispatched lay first responders), proportion of initial shockable rhythm, number of patients with intervention, and survival to hospital discharge/30-day survival in percentages. In case the studies had divided survival into subgroups, we combined these to generate the overall survival, by using the formula for combining groups from the Cochrane handbook.²⁷

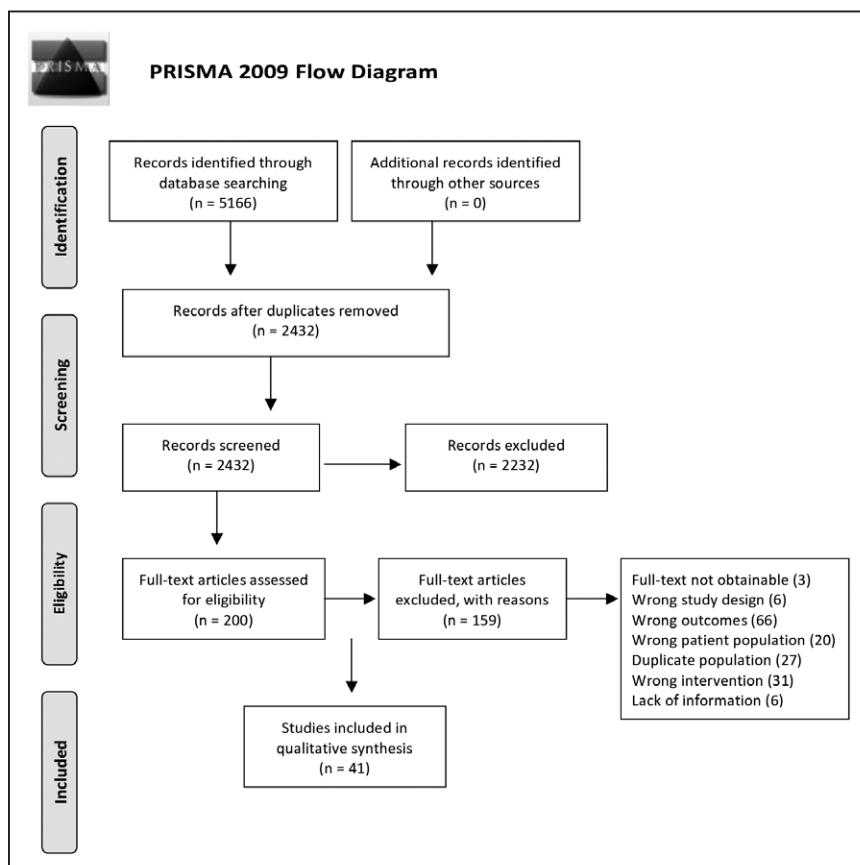


Figure 2. The PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analysis) flow-chart of the study selection.

Risk of Bias in Individual Studies

Risk of bias was assessed by using the Newcastle Ottawa Scale, which is developed specifically for nonrandomized interventional studies and comparative studies.²⁸ A study can be awarded from 0 to 9 stars, with 9 stars representing a low risk of bias, and 0 stars a high risk of bias. Stars are awarded according to representativeness of the exposed cohort (maximum 2 stars), ascertainment of exposure (maximum 1 star), assessment of outcome adequacy of follow-up for cohorts (maximum 1 star), demonstration that the outcome of interest was not present at the start of the study (maximum 1 star), follow-up being long enough for outcomes to occur (maximum 1 star), selection of the non-exposed (maximum 1 star), and comparability of cohorts on the basis of the design or analysis (maximum 2 stars). The 2 last-mentioned points could never be awarded to our included studies as per the study design, and thus our studies could be awarded a maximum of 6 stars. The stars serve as an indicator of the quality of the specific studies and can

serve as guidance for the readers to compare the reported results between studies.

The median and range of Newcastle Ottawa Scale values are presented for the included studies.

Statistical Analysis

The median survival to hospital discharge was calculated for each subgroup. Because of the pronounced heterogeneity in the subgroups, it was impossible to perform meaningful meta-analyses.²⁹ The calculations and figures were performed by using RStudio Version 0.99.486.

RESULTS

Through our literature search, we identified 5166 records. After the removal of duplicates and nonoriginal research, 2432 studies were assessed for inclusion through title and abstract screening. Here, 200 studies

Table 1. Study Characteristics: Nondispatched Lay First Responders (On-Site Automated External Defibrillator)

Author, Year of Publication	Study Period	Study Duration in Months	Geographical Location	Study Design	Name of Registry/Data Source	Newcastle Ottawa Scale
O'Rourke et al, 1997 ⁴²	N/A	65	Australia (airport and airline)	Prospective	N/A	6
Page et al, 2000 ⁴⁴	1997–1999	25.5	United States (airline)	N/A	Medical department of American Airlines and review of ECGs	6
Valenzuela et al, 2000 ¹¹	1997–1999	32	United States	Prospective	Data provided by Clark County Fire Department	6
Caffrey et al, 2002 ³¹	1999–2001	24	United States, Illinois (airports)	Prospective	Incident reports, paramedic records, telephone interviews	6
Fedoruk et al, 2002 ³⁴	1994–1999	60	United States	Retrospective	Ambulance call reports, fire service medical assist reports, casino's medical records, data from security cameras	6
Bertrand et al, 2004 ³⁰	2002–2003	12	France (Air France)	N/A	Specialists analyzed ECGs from AEDs	6
Chan et al, 2007 ⁴³	1996–2004	106	Australia	N/A	N/A	6
Colquhoun et al, 2008 ¹⁸	1999–2005	82	United Kingdom	N/A	N/A	6
Weisfeldt et al, 2010 ²²	2005–2007	17	United States	Prospective	ROC Epistry Cardiac Arrest Registry	6
Berdowski et al, 2011 ¹⁰	2006–2009	39	Netherlands	Retrospective	ARREST	6
McNally et al, 2011 ³⁹	2005–2010	63	United States	N/A	CARES	6
Eckstein, 2012 ³³	2000–2012	120	United States, California	Prospective	Reviewers of AED use and prehospital care report	6
Drezner et al, 2013 ³²	2009–2011	24	United States	Prospective	National Registry for AED Use in Sports	5
Jorgenson et al, 2013 ³⁵	2002–2009	85	United States	Prospective	Reports by patient, telephone follow-up interviews by physician	4
Maron et al, 2013 ³⁸	1970–2012	504	United States, Minnesota	Pro- and retrospective	US Commotio Cordis Registry	3
Wissenberg et al, 2013 ⁷	2001–2010	115	Denmark	Retrospective	Danish Cardiac Arrest Registry and Danish National Patient Registry	6
Hansen et al, 2014 ⁴⁰	2011–2013	20	Denmark	Prospective	Danish Cardiac Arrest Registry and medical records	6
Kilaru et al, 2014 ³⁶	1999–2012	168	United States	Retrospective	FOH AED Program database	6
Lijovic et al, 2014 ³⁷	2002–2013	132	Australia	Retrospective	VACAR	6
Nakahara et al, 2015 ⁴¹	2005–2012	96	Japan	Retrospective	All Japan Utstein Registry	6
Ringh et al, 2015 ⁶³	2006–2012	84	Sweden	Retrospective	SCAR	6

AED indicates automated external defibrillator; ARREST, The Amsterdam Resuscitation Study; CARES, Cardiac Arrest Registry to Enhance Survival; FOH, Federal Occupational Health; N/A, not available; ROC, Resuscitation Outcomes Consortium; SCAR, The Swedish Cardiac Arrest Register; and VACAR, Victorian Ambulance Cardiac Arrest Registry.

were included with a Cohen's κ of 0.76 (95% confidence interval, 0.71–0.81; 2-sided P value<0.001). Full text eligibility of the remaining studies was assessed, of which 159 studies were excluded, leaving 41 studies from 11 different countries in our review (Figure 2).

The median study duration was 40.5 months (range, 10–504). All studies were either prospective or retrospective observational studies. Study characteristics are summarized in Tables 1 and 2.

Of the 41 included studies, 18 reported nondispatched lay first responder AED usage,^{7,11,18,30–44} and 20 studies reported EMDC-dispatched professional lay first responder AED usage.^{13,14,45–62} We identified 3 studies that reported both groups.^{10,22,63} We did not identify any studies reporting AED usage by the third subgroup, EMDC-dispatched lay first responders. The individual study outcomes are summarized in Tables 3 and 4.

Three of the studies reported an outcome of 30-day survival,^{7,40,63} whereas the remaining reported survival to hospital discharge. The median proportion of survivors after OHCA where AEDs were reported to be attached (and the defibrillation status was unknown or not specified) by non-EMS was 22.3% (range, 4.0–78.0). The median proportion of survivors after OHCA where an AED was confirmed to have defibrillated the patient was 40.0% (range, 9.0–76.0).

When the 2 subgroups were examined separately, we found the median proportion of survivors to be 32.0% (range, 14.0–78.0) when nondispatched lay first responders had attached the AED, and 11.5% (range, 4.0–21.1) when EMDC-dispatched professional first responders (firefighters/police) had attached the AED.

It is notable that when an AED was confirmed to have defibrillated the patient with OHCA, the median proportion of survivors to hospital discharge was 53.0% (range, 26.0–72.0) for the subgroup of nondispatched lay first responders, and 28.6% (range, 9.0–76.0) for EMDC-dispatched professional first responders (firefighters/police).

The results are illustrated in Table 5. Information on the cerebral performance category scores of the survivors was extremely limited but can be found in [Tables I and II in the online-only Data Supplement](#).

The risk of bias between the studies showed our highest possible median Newcastle Ottawa Scale score of 6 (range, 3–6). Only 6 of the studies were awarded a score <6 (4 studies received 5 stars, 1 received 4, and 1 received only 3 stars) indicating the overall high quality of our included studies.

DISCUSSION

Summary of Evidence

In this systematic review of observational studies, we aimed at evaluating the effect on survival to hospital discharge for patients with OHCA according to type

of PAD strategy implemented. Our review showed a median survival to hospital discharge after OHCA of 40% when patients were defibrillated before EMS arrival by PAD. Furthermore, survival after OHCA seemed to be higher when the patient was defibrillated by nondispatched lay first responders in comparison with EMDC-dispatched professional first responders (firefighters/police), with a median survival to hospital discharge of 53.0% (range, 26–72) and 28.6% (range, 9–76), respectively. We did not identify any studies describing long-term survival after defibrillation performed by EMDC-dispatched lay first responders.

PAD is recommended by the 2015 American Heart Association Guidelines and the 2015 European Resuscitation Council Guidelines, as well.^{8,15} The recommendations are based on studies demonstrating increased survival after OHCA following defibrillation with an AED before the arrival of EMS.¹¹ However, the guidelines do not distinguish between the different types of PAD strategies. Our review found that different PAD strategies were associated with much higher survival proportions than those reported for OHCA treated by EMS only, which average $\approx$ 10% to 12%.^{64,65} Regardless of the PAD strategy, this may be explained by a reduced time from collapse to CPR and defibrillation with an AED,^{66,67} because the chance of survival decreases by 7% to 10% for every minute that passes without intervention.⁶

Accordingly, the better outcome of PAD by nondispatched lay first responders (=on-site AED) in comparison with PAD by EMDC-dispatched professional first responders (=mobile AED), may also be explained by a faster on-scene arrival, and thus a decreased time to defibrillation. A recent Dutch study supports this, because it shows nondispatched lay first responders shock patients with OHCA after a median of 4:06 minutes (interquartile range, 2:23–6:40 minutes), whereas EMDC-dispatched professional first responders (firefighters/police) shock the patients after a median of 8:03 minutes (interquartile range, 6:43–10:13).⁶⁸ The Public Access Defibrillation Trial also supports training and equipping volunteers to attempt early defibrillation within a structured response system, and found that doing so may double the number of survivors to hospital discharge after OHCA.⁶⁹

Our review did not identify any studies describing defibrillation performed by EMDC-dispatched lay first responders. However, focus on EMDC-dispatched lay first responders is rapidly developing because it is thought to potentially cover a large population. This would include arrests in residential areas, where 70% of OHCA occur,⁷² and where the effect of on-site AEDs is limited.^{69,71} A recent randomized clinical trial from Sweden showed increased rates of bystander

Table 2. Study Characteristics: Emergency Dispatch Center–Dispatched Professional First Responders (Police/Firefighter) (Mobile Automated External Defibrillator)

Author, Year of Publication	Study Period	Study Duration in Months	Geographical Location	Study Design	Name of Registry/ Data Source	Newcastle Ottawa Scale
Weaver et al, 1984 ⁵⁴	1980–1982	24	United States, Washington	Prospective	Interviews and hospital records	6
Weaver et al, 1988 ⁵⁶	1984–1986	25	United States, Washington	Prospective	Paramedic and hospital records	6
Callahan et al, 1993 ⁶⁰	1989–1990	23	United States, California	Prospective	Data from fire department, paramedics, and emergency department inpatient records	6
Weaver et al, 1993 ⁵⁵	N/A	24	United States, Washington	N/A	AED recordings and prehospital and hospital records	6
Ho et al, 1997 ⁶²	1993–1995	30	United States, Minnesota	Retrospective	MECC computer data bank and the fire department's annual report	5
Mosesso et al, 1998 ⁴⁹	1992–1995	36	United States, Pennsylvania	Prospective	Police- and paramedic reports	5
Groh et al, 2001 ⁴⁶	1997–1999	10	United States, Indiana	Pro- and retrospective	PARADE	6
Capucci et al, 2002 ⁵⁹	1999–2000	22	Italy	Prospective	Emergency Department database and AED information	6
Forrer et al, 2002 ⁴⁵	N/A	120	United States, Michigan	Retrospective	William Beaumont Hospital cardiac arrest registry	6
MacDonald et al, 2002 ⁴⁸	1995–1999	60	United States, Massachusetts	Prospective	Boston EMS cardiac arrest quality assurance database	6
Carpenter et al, 2003 ⁶¹	1999–2002	44	United States, Washington	Retrospective	AED records, EMS records, hospital records	6
Persse et al, 2003 ⁵⁰	1997	12	United States	Retrospective	Patient records	6
Van Alem et al, 2003 ⁵⁸	2000–2002	28	Netherlands	Prospective	AED data, interviews	6
Valenzuela et al, 2005 ⁵²	2001–2002	12	United States, Arizona	Retrospective	AEDs and Tucson Fire Department cardiac arrest database	6
White et al, 2005 ⁵³	1990–2003	157	United States, Minnesota	N/A	AEDs and the Public Safety Communications Center	6
Høyer and Christensen, 2009 ¹⁴	2005–2007	28	Denmark	Retrospective	The National Emergency Call Center, the EMS, and the Fire Department	5
Weisfeldt et al, 2010 ²²	2005–2007	17	United States	Prospective	ROC Epistry Cardiac Arrest Registry	6
Berdowski et al, 2011 ¹⁰	2006–2009	39	Netherlands	Retrospective	ARREST	6
Becker et al, 2013 ¹³	2010–2012	36	United States, Washington	Prospective	Registry King County Emergency Medical Services Division + Police Reports	6
Saner et al, 2013 ⁵¹	2001–2008	96	Switzerland	Prospective	Record sheets from EMS and first-responders and recordings from AED devices	6
Hansen et al, 2015 ⁴⁷	2010–2013	48	United States, North Carolina	Prospective	CARES	6
Ringh et al, 2015 ⁶³	2006–2012	84	Sweden	Retrospective	SCAR	6
Telesz et al, 2015 ⁵⁷	2006–2013	96	United States, Minnesota	Retrospective	Hospital charts and information from the Emergency Call Center	6

AED, automated external defibrillator; ARREST, The Amsterdam Resuscitation Study; CARES, Cardiac Arrest Registry to Enhance Survival; EMS, emergency medical services; MECC, Minneapolis Emergency Communication Center; N/A, not available; PARADE, The Police as Responder Automated Defibrillation evaluation; and ROC, Resuscitation Outcomes Consortium.

CPR and shorter time to defibrillation in comparison with EMS, when volunteers were activated via a mobile phone–positioning system and sent to OHCA within 500 m of them.¹² In the Netherlands, comparable effects were seen in a study where mobile phone text messages sent lay responders within 1000 m of patients with suspected OHCA to per-

form bystander CPR and use AEDs.⁶⁸ In the United States and the United Kingdom, similar text message systems are appearing.^{72,73} Engaging laypersons to either use an on-site AED or dispatch them via the EMDC thus seems especially promising, because, not only can they reach the patient with OHCA more quickly, but they can also reach patients with OHCA

Table 3. Outcomes: Nondispatched Lay First Responders (On-Site Automated External Defibrillator)

Author, Year of Publication	Proportion of Bystander Cardiopulmonary Resuscitation, %			Proportion of Witnessed OHCA's, %			Initial Shockable Rhythm, %	No. of Patients With Intervention		Survival to Hospital Discharge/30-Day Survival, %	
	All OHCA's	AED Attached	AED Defibrillated	All OHCA's	AED Attached	AED Defibrillated		AED Attached	AED Defibrillated	AED Attached	AED Defibrillated
O'Rourke et al, 1997 ⁴²	–	–	–	–	76.0	–	50.0	46	23	–	26.0
Page et al, 2000 ⁴⁴	–	–	–	–	–	–	–	–	15	–	40.0*
Valenzuela et al, 2000 ¹¹	–	43.0	–	–	–	–	71.0	148	105	38.0	53.0*
Caffrey et al, 2002 ³¹	–	–	94.0	–	91.0	100	82.0	22	18	–	56.0*
Fedoruk et al, 2002 ³⁴	100	100	100	100	100	100	86.0	23	18	65.0	–
Bertrand et al, 2004 ³⁰	–	–	–	–	75.0	–	42.0	12	5	17.0	40.0*
Chan et al, 2007 ⁴³	17.0	–	–	–	–	–	31.0	35	11	14.0	45.0
Colquhoun et al, 2008 ¹⁸	–	–	64.0	–	–	57.6	79.0	437	347	25.9	30.5
Weisfeldt et al, 2010 ²²	–	–	–	–	–	–	–	101	–	40.0	–
Berdowski et al, 2011 ¹⁰	–	99.0	–	–	90.0	–	76.0	128	97	49.2*	63.9*
McNally et al, 2011 ³⁹	33.3	–	–	47.3	–	–	–	1166	–	23.5	–
Eckstein, 2012 ³³	–	–	–	–	–	–	66.0	59	39	–	69.0*
Drezner et al, 2013 ³²	92.0	–	–	93.0	–	–	–	41	–	78.0	–
Jorgenson et al, 2013 ³⁵	–	88.0	–	–	76.0	–	56.0	25	14	32.0	57.0
Maron et al, 2013 ³⁸	–	–	–	100	–	–	–	–	16	–	69.0
Wissenberg et al, 2013 ⁷	30.8	–	–	52.0	–	–	–	241	–	27.8	–
Hansen et al, 2014 ⁴⁰	–	92.0	98.0	–	66.0	92.0	55.0	91	50	42.0	72.0*
Kilaru et al, 2014 ³⁶	–	–	–	–	70.0	–	72.0	96	65	25.0	–
Lijovic et al, 2014 ³⁷	–	–	95.0	–	–	88.0	–	–	153	–	45.0
Nakahara et al, 2015 ⁴¹	47.0	–	–	100	–	–	–	–	2287	–	40.7
Ringh et al, 2015 ⁶³	–	–	62.0	54.0	–	–	–	–	74	–	70.0*

Dashes indicate that data could not be extracted. AED indicates automated external defibrillator; and OHCA, out-of-hospital cardiac arrests.

*Studies that reported Cerebral Performance Scores. Further details can be found in Table 1 in the online-only Data Supplement.

in residential areas. At present, studies demonstrating this subgroup's effect on survival after OHCA are lacking, but a randomized control trial from Korea with this outcome is under conduction.⁷⁴ However, it is noteworthy that the focus of this review was not the means of providing access to AEDs.

Strengths and Limitations at Study and Outcome Level

Definitions of the terms PAD and first responder lack consistency throughout the included studies. Some understand PAD to be a possibility solely for laypersons,

Table 4. Outcomes: Emergency Medical Dispatch Center–Dispatched Professional First Responders (Police/Firefighter) (Mobile-AED)

Author, Year of Publication	Proportion of Bystander Cardiopulmonary Resuscitation, %			Proportion of Witnessed OHCA, %			Initial Shockable Rhythm, %	No. of Patients With Intervention		Survival to Hospital Discharge/30-Day Survival, %	
	All OHCA	AED Attached	AED Defibrillated	All OHCA	AED Attached	AED Defibrillated		AED Attached	AEDs Defibrillated	AED Attached	AED Defibrillated
Weaver et al, 1984 ⁵⁴	–	–	30.0	–	–	60.0	–	–	87	–	30.0
Weaver et al, 1988 ⁵⁶	–	–	–	–	–	79.0	40.0	687	242	–	30.5
Callahan et al, 1993 ⁶⁰	–	16.0	28.0	–	53.0	79.0	33.0	835	274	4.0	9.0
Weaver et al, 1993 ⁵⁵	–	–	45.0	–	–	72.0	34.0	847	289	–	26.3
Ho et al, 1997 ⁶²	–	75.6	–	–	–	–	30.4	358	109	–	20.2*
Mosesso et al, 1998 ⁴⁹	–	–	–	66.0	–	–	39.0	118	46	–	26.1
Groh et al, 2001 ⁴⁶	–	73.1	–	–	65.4	–	77.0	26	20	11.5	15.0
Capucci et al, 2002 ⁵⁹	–	–	–	73.7	–	–	23.8	143	34	10.5	44.1*
Forrer et al, 2002 ⁴⁵	–	21.9	–	–	67.3	–	50.0	128	62	11.9	14.8
MacDonald et al, 2002 ⁴⁸	26.3	–	–	86.8	–	–	77.8	36	28	21.1	28.6
Carpenter et al, 2003 ⁶¹	59.3	–	–	–	–	77.3	–	–	366	–	28.7
Persse et al, 2003 ⁵⁰	–	–	–	34.2	–	–	–	–	61	–	19.7
Van Alem et al, 2003 ⁵⁸	–	40.0	–	–	79.0	–	52.0	184	96	8.7	16.6
Valenzuela et al, 2005 ⁵²	–	52.0	–	–	51.0	–	34.0	61	20	7.0	–
White et al, 2005 ⁵³	–	–	48.0	63.5	–	82.0	–	–	100	–	43.0*
Høyer and Christensen, 2009 ¹⁴	–	–	–	–	–	–	28.0	29	8	–	75.0
Weisfeldt et al, 2010 ²²	–	–	–	–	–	–	–	75	–	13.0	–
Berdowski et al, 2011 ¹⁰	–	89.0	–	–	74.0	–	50.0	478	238	17.2*	32.4*
Becker et al, 2013 ¹³	–	–	–	–	–	–	29.0	21	6	–	76.0
Saner et al, 2013 ⁵¹	–	–	–	–	–	–	15.0	619	93	–	14.0*
Hansen et al, 2015 ⁴⁷	45.7	–	–	44.8	–	–	–	–	774	–	25.2
Ringh et al, 2015 ⁶³	–	–	34.0	54.0	–	–	–	–	53	–	42.0
Telesz et al, 2015 ⁵⁷	–	–	62.8	100	–	–	–	–	108	–	57.4*

Dashes indicate that data could not be extracted. AED indicates automated external defibrillator; and OHCA, out-of-hospital cardiac arrest.

*Studies reporting Cerebral Performance Scores. Further details can be found in Table II in the online-only Data Supplement.

Table 5. Studies Reporting Out-of-Hospital Cardiac Arrest Survival for Automated External Defibrillator Attached and Automated External Defibrillator Defibrillated According to Type of Automated External Defibrillator User

AED Intervention*	User†	Median Survival, %‡	Range	Interquartile Range	No. of Studies
AED attached	Both	22.3	4.0–78.0	0.12–0.37	22
AED attached	Police/firefighter	11.5	4.0–21.1	0.09–0.13	9
AED attached	Layperson	32.0	14.0–78.0	0.25–0.42	13
AED defibrillated	Both	40.0	9.0–76.0	0.26–0.56	36
AED defibrillated	Police/firefighter	28.6	9.0–76.0	0.20–0.42	21
AED defibrillated	Layperson	53.0	26.0–72.0	0.26–0.56	15

The table shows the actual number of studies reporting survival status for AED attachment or AED defibrillation according to users among all the included studies (n=41). AED indicates automated external defibrillator; and OHCA, out-of-hospital cardiac arrests.

*AED attached: the AED was either only attached or not specified to have defibrillated the patient with OHCA. AED defibrillated: the AED was confirmed to have defibrillated the patient.

†Both indicates nondispatched lay first responders and professional first responders (firefighters/police) dispatched by the Emergency Medical Dispatch Center; police/firefighter, professional first responders (firefighters/police) dispatched by the Emergency Medical Dispatch Center; and layperson, nondispatched lay first responders.

‡Median survival to hospital discharge or 30-day survival.

whereas others understand it to include firefighters and police. The term first responder is understood to include, or exclude, EMS personnel with variation. These confusions are partly because of definitions changing over time and the lack of consensus on the terms. Whenever the definitions used for PAD and first responders were impossible to interpret, the studies were excluded.

Furthermore, to ensure equal etiologies of the cardiac arrests, we excluded studies that only reported survival of patients <18 years of age.

Because the majority of studies failed to specify whether AEDs were only attached to the patient with OHCA (eg, because of a nonshockable rhythm) or delivered a shock, we categorized it as only attached in case it was not clear. As a result, it is assumed that the studies in this group include a high degree of heterogeneity.²⁹ Furthermore, the inclusion criteria among the included studies, eg, in terms of witness status, were found to be highly diverse. For these 2 reasons, it was not possible to undertake a meta-analysis.

Strengths and Limitation at Review Level

Our review is conducted according to the PRISMA guidelines guaranteeing a uniform and systematic approach, and transparency of the methods, as well.²³ Studies from a variety of countries are included, creating a broad perspective. To ensure homogeneity in methodology, only observational studies are included in our review. Survival to hospital discharge and 30-day survival are clearly 2 different outcomes. However, because the ease of collecting information within different healthcare systems varies, the Utstein recommendations accept both survival to hospital discharge and 30-day survival as survival end points.⁷⁵ When comparing cardiac arrest survival across several studies, it there-

fore seems inevitable not to include both forms of survival reporting. We did, however, exclude studies only reporting return of spontaneous circulation to ensure an estimation of the long-term effect of PAD.

Furthermore, only studies written in English were included, and studies containing <10 cases of OHCA were excluded to weaken potential publication bias.

Only a few of the included studies evaluate the cerebral performance category score of the survivors, and this outcome could therefore not be fully explored despite its importance (Tables I and II in the online-only Data Supplement). However, a recent Danish study showed that 76.6% of survivors of OHCA, who were employed before their cardiac arrest, returned to work, suggesting that the majority of survivors to hospital discharge in the included studies obtain a good cerebral outcome.⁷⁶ Finally, studies describing the group of lay first responders appear to be of newer date than studies describing the group of professional first responders. Although newer AED devices and more knowledge may improve survival independently, we chose not to exclude the older studies, because a sensitivity analysis showed only a minor improvement in survival for the professional first responders of 2.2% points when the year of publication was taken into account.

CONCLUSION

This systematic review reports a median survival to hospital discharge of 40% for patients with OHCA defibrillated before the arrival of EMS by PAD programs. In addition, significantly higher survival was associated with defibrillation by nondispatched lay first responders in comparison with EMDC-dispatched professional first responders (firefighters/police). New methods of AED

recruitment by EMDC-dispatched lay first responders, eg, via mobile phone positioning and dispatch-aided AED retrieval might be effective, but studies reporting the effect on survival proportions are still lacking.

SOURCES OF FUNDING

The authors received an unrestricted grant from the Danish foundation TrygFonden for the work of this publication. TrygFonden had no involvement in creating the study, the creative process, or the evaluation of the results. The research group at the Emergency Medical Services, Copenhagen receives Center-support from the Laerdal-foundation for acute medicine.

DISCLOSURES

None.

AFFILIATIONS

From Emergency Medical Services Copenhagen, University of Copenhagen, Denmark (J.S.B., S.V., T.P.M., F.L., F.F.); and National Institute of Public Health, University of Southern Denmark, Copenhagen (A.K.E.).

FOOTNOTES

Received April 20, 2017; accepted June 7, 2017.

The online-only Data Supplement is available with this article at <http://circ.ahajournals.org/lookup/suppl/doi:10.1161/CIRCULATIONAHA.117.029067/-DC1>.

Circulation is available at <http://circ.ahajournals.org>.

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